Problem Set 16

Triangle Substitution

October 13, 2026

Problem 6

(a)

$$\int \frac{x}{\sqrt{9-x^2}}\,dx = \begin{bmatrix} u = 9-x^2 \\ du = -2x\,dx \end{bmatrix} -\frac{1}{2}\int \frac{du}{\sqrt{u}}$$
$$= -\frac{1}{2}\cdot 2\sqrt{u} = \boxed{-\sqrt{9-x^2}+C}$$

(b)

$$\int \frac{x^3}{\sqrt{9-x^2}}\,dx = \begin{bmatrix} u = 9-x^2 \\ du = -2x\,dx \end{bmatrix} \frac{1}{2}\int \frac{u-9}{\sqrt{u}}\,du$$
$$= \frac{1}{2}\int \sqrt{u} - \frac{9}{\sqrt{u}}\,du = -9\sqrt{u} + \frac{1}{3}u^{\frac{3}{2}}$$
$$= \boxed{\frac{1}{3}(9-x^2)^{\frac{3}{2}} - 9\sqrt{9-x^2}+C}$$

(c)

$$\int \sqrt{9-x^2}\,dx = \int \sqrt{9-(3\sin\theta)^2}3\cos x\,d\theta$$
$$= 9\int \cos^2\theta\,d\theta = \frac{9}{2}\int \cos 2\theta + 1\,d\theta$$
$$= \frac{9}{2}\sin\theta\cos\theta + \frac{9\theta}{2} + C$$
$$= \boxed{\frac{9}{2}\arcsin\frac{x}{3} + \frac{x\sqrt{9-x^2}}{2} + C}$$

(d)

$$\int \frac{dx}{9+x^2} = \begin{bmatrix} u = \frac{x}{3} \\ du = \frac{1}{3}\,dx \end{bmatrix} \frac{1}{3}\int \frac{1}{1+u^2}\,du$$
$$= \frac{1}{3}\arctan u + C = \boxed{\frac{1}{3}\arctan\frac{x}{3} + C}$$

Problem 7

(b)

$$\begin{aligned}\int \frac{dx}{(x^2-4x+13)^{\frac{3}{2}}} &= \int \frac{dx}{((x-2)^2+9)^{\frac{3}{2}}} \\ &= \begin{bmatrix} u = x-2 \\ du = dx \end{bmatrix} \int \frac{du}{(u^2+9)^{\frac{3}{2}}} \\ &= \int \frac{3\sec^2\theta\, d\theta}{27\sec^3\theta} = \frac{1}{9}\int \cos\theta\, d\theta \\ &= \frac{1}{9}\sin\theta + C = \frac{1}{9}\cdot\frac{u}{\sqrt{u^2+9}} + C \\ &= \boxed{\frac{x-2}{9\sqrt{x^2-4x+13}} + C}\end{aligned}$$

Problem 9

(a)

$$\sin\frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}},\ \cos\frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$

$$\sin\theta = 2\cdot\frac{t}{\sqrt{1+t^2}}\cdot\frac{1}{\sqrt{1+t^2}} = \boxed{\frac{2t}{1+t^2}}$$

$$\cos\theta = \left(\frac{1}{\sqrt{1+t^2}}\right)^2 - \left(\frac{t}{\sqrt{1+t^2}}\right)^2 = \boxed{\frac{1-t^2}{1+t^2}}$$

$$\theta = 2\arctan t \implies \boxed{d\theta = \frac{2\,dt}{1+t^2}}$$

(b)

$$\int \frac{d\theta}{2+\cos\theta} = \int \frac{2dt}{1+t^2}\cdot\frac{1+t^2}{3+t^2} = \int \frac{2}{t^2+3}\,dt$$

$$= \frac{2}{\sqrt{3}}\arctan\frac{t}{\sqrt{3}} + C = \boxed{\frac{2}{\sqrt{3}}\arctan\left(\frac{1}{\sqrt{3}}\tan\frac{\theta}{2}\right) + C}$$

(c) It is impossible, as the integrand is always positive. The mistake is due to the domain restriction of the half-angle formulas.

$$\int_0^{2\pi} \frac{d\theta}{2+\cos\theta} = \lim_{x\to\pi^-} F(x) - \lim_{x\to-\pi^+} F(x) = 2\cdot\frac{\pi}{\sqrt{3}} = \boxed{\frac{2\pi}{\sqrt{3}}}$$

(d)

$$a > b \implies \frac{1}{a+b\cos\theta} > 0$$

$$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \int_{-\pi}^{\pi} \frac{d\theta}{a+b\cos\theta}$$
$$= \int_{-\infty}^{\infty} \frac{2\,dt}{1+t^2} \cdot \frac{1+t^2}{(a+b)+(a-b)t^2}$$
$$= \int_{-\infty}^{\infty} \frac{2\,dt}{(a+b)+(a-b)t^2} = \frac{2}{a-b}\int_{-\infty}^{\infty} \frac{dt}{\frac{a+b}{a-b}+t^2}$$
$$= \begin{bmatrix} u = t\sqrt{\frac{a-b}{a+b}} \\ du = dt\sqrt{\frac{a-b}{a+b}} \end{bmatrix} \frac{2}{a-b}\sqrt{\frac{a-b}{a+b}} \left[\arctan t\sqrt{\frac{a-b}{a+b}}\right]_{-\infty}^{\infty}$$
$$= \boxed{\frac{2\pi}{\sqrt{a^2-b^2}}}$$